Problem 8

Evaluate the integral:

$$I = \int \frac{\sqrt{x^2 + 2x + 1}}{x} \, dx.$$

Solution

To solve this integral, we first simplify the square root expression and then proceed to integrate step by step.

Step 1: Simplify the square root

Observe that:

$$x^2 + 2x + 1 = (x + 1)^2.$$

Thus, the integral becomes:

$$I = \int \frac{\sqrt{(x+1)^2}}{x} \, dx.$$

Since the square root of $(x+1)^2$ is $|x+1|$, we write:

$$I = \int \frac{|x+1|}{x} \, dx.$$

Step 2: Handle the absolute value function

The absolute value $|x+1|$ depends on the sign of $x+1$:

$$|x+1| = \begin{cases} x+1, & \text{if } x \geq -1; \\ -(x+1), & \text{if } x < -1. \end{cases}$$

We split the integral into two cases: $x \geq -1$ and $x < -1$.

Case 1: $x \geq -1$

For $x \geq -1$, $|x+1| = x+1$. Substituting this into the integral:

$$I_1 = \int \frac{x+1}{x} \, dx.$$

Simplify the fraction:

$$I_1 = \int \left(1 + \frac{1}{x}\right) dx.$$

Integrating each term:

$$I_1 = \int 1 \, dx + \int \frac{1}{x} \, dx = x + \ln|x| + C_1.$$

Case 2: $x < -1$

For $x < -1$, $|x+1| = -(x+1)$. Substituting this into the integral:

$$I_2 = \int \frac{-(x+1)}{x} \, dx = -\int \left(1 + \frac{1}{x}\right) dx.$$

Simplify and integrate:

$$I_2 = -\int 1 \, dx - \int \frac{1}{x} \, dx = -x - \ln|x| + C_2.$$

Final Answer

Combining the two cases, the solution to the integral is:

$$I = \begin{cases} x + \ln|x| + C_1, & \text{if } x \geq -1; \\ -x - \ln|x| + C_2, & \text{if } x < -1. \end{cases}$$